MTH 254 final exam — coverage map

Every objective from the study guide, and where the notes cover it.

Sections 2.1 & 2.2: 2D and 3D Vectors

1. Addition, subtraction, scalar multiplication — vector-calc p1
2. Unit vector in the same direction — vector-calc p1
3. Magnitude of a vector — vector-calc p1
4. Distance between two points — vector-calc p1

Section 2.3: Dot Products

5. Dot product, components and angle definitions — vector-calc p1
6. Angle between two vectors — vector-calc p1
7. Determine when two vectors are orthogonal — vector-calc p1
8. Orthogonal projection of one vector onto another — vector-calc p1

Section 2.4: Cross Products

9. Compute the cross product — vector-calc p2
10. Vector orthogonal to two given vectors — vector-calc p2
11. Area of the triangle between three points — vector-calc p2
12. Area of a parallelogram, from points or vectors — vector-calc p2

Section 2.5: Lines and Planes

13. Line through two given points — vector-calc p4
14. Line parallel to a vector through a point — vector-calc p4
15. Plane through three given points — vector-calc p4
16. Plane through a point, parallel to two vectors — vector-calc p4
17. Plane through a point, perpendicular to a vector — vector-calc p4
18. Line of intersection of two planes — vector-calc p4
19. Identify a normal vector from a plane equation — vector-calc p4

Section 3.1: Vector-Valued Functions

20. Understand how vector-valued functions work — vector-calc p7
21. Identify curves from their equations — vector-calc p7
22. Point of intersection of two vector functions — vector-calc p7

Section 3.2: Calculus of Vector-Valued Functions

23. Derivative of a vector-valued function — vector-calc p8
24. Unit tangent vector at a point — vector-calc p8
25. Derivative of dot and cross products — vector-calc p8
26. Integral, definite and indefinite — vector-calc p8
27. Draw or choose the tangent vector at t=a — vector-calc p8

Section 3.3: Arc Length

28. Set up an integral for the length of a curve — vector-calc p10
29. Calculate the length of a given curve — vector-calc p10

Section 3.4: Motion in Space

30. From a(t), v(0), s(0) find v(t) and s(t) — vector-calc p11
31. Motion in a gravitational field — vector-calc p11
32. Interpret a contextual motion situation — vector-calc p11

Section 4.1: Graphs and Level Curves

33. Match graphs of surfaces to their equations — vector-calc p6
34. Create level curve diagrams — vector-calc p12

Section 2.6: Cylinders and Quadric Surfaces

35. Identify a cylinder from its graph — vector-calc p6
36. Identify a cylinder from its formula — vector-calc p6
37. Identify quadric surfaces from their graphs — vector-calc p6
38. Identify quadric surfaces from their formulas — vector-calc p6

Section 4.2: Limits and Continuity

39. Evaluate multivariable limits — vector-calc p13
40. Two-path test for a limit that does not exist — vector-calc p13
41. Continuity of a multivariable function — vector-calc p13

Section 4.3: Partial Derivatives

42. Find partial derivatives — vector-calc p14
43. Value of a partial derivative at a point — vector-calc p14
44. Interpret partial derivative values — vector-calc p14
45. Second partial derivatives — vector-calc p14
46. Clairaut's theorem — vector-calc p14

Section 4.5: The Chain Rule

47. Multivariable chain rule, functions and values — vector-calc p16
48. Interpret chain rule values — vector-calc p16

Section 4.6: Directional Derivatives and the Gradient

49. Directional derivatives in context — vector-calc p18
50. Interpret a directional derivative value — vector-calc p18
51. The gradient and its link to directional derivs — vector-calc p18
52. Directions of fastest change and no change — vector-calc p18
53. Gradient against the level curves of a surface — vector-calc p18

Section 4.4: Tangent Planes and Linear Approximation

54. Tangent plane equations for surfaces — vector-calc p15
55. Linear approximation of function values — vector-calc p15

Section 4.7: Maximum/Minimum Problems

56. Find critical points — vector-calc p20
57. Classify with the Second Derivative Test — vector-calc p20
58. Know what a saddle point is — vector-calc p20
59. Absolute extrema on a closed bounded set — vector-calc p20

Chapter 5 (General)

60. Decide which coordinate system to use — vector-calc p30
61. Average value of a function over a region — vector-calc p24
62. Evaluate double and triple integrals — vector-calc p23
63. Volume between two surfaces — vector-calc p25
64. Rearrange the order of a triple integral — vector-calc p27
65. Integrate f=1 for area and for volume — vector-calc p24
66. How dA and dV work in every system — vector-calc p29

Sections 5.1 and 5.2: Double Integrals, Cartesian

67. Set up double integrals over a region — vector-calc p24
68. Reverse the order of integration — vector-calc p24

Section 5.3: Double Integrals, Polar

69. Convert between Cartesian and polar — vector-calc p29

Section 5.4: Triple Integrals, Cartesian

70. Set up triple integrals for volume and mass — vector-calc p27

Section 5.5: Triple Integrals, Cylindrical and Spherical

71. Convert between Cartesian, cylindrical, spherical — vector-calc p29